Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Three-Dimension Two-Photon Endomicroscopy for Biopsy-free Real-time Optical Histology in vivo			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio・Medical		**Electrical・Electronics**	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Optical Application Devices	Imaging Devices
Concept of Proposed Technology	**Core Concept:** The proposed two-photon endomicroscopy (2PEM) technology represents a new paradigm for clinical diagnosis and potentially for guidance of intervention. 2PEM enables **optical histology *in vivo*** by using nonlinear excitation to generate intrinsic tissue contrast (e.g., based on autofluorescence and second-harmonic generation) at cellular and subcellular resolution, without the need for tissue removal. By confining excitation to the focal volume and using near-infrared light, it achieves optical sectioning deep within scattering tissue while minimizing photodamage. **Main Functions:** 2PEM is implemented using flexible fiber-optic probes with a small footprint (typically 2–4 mm in diameter). It can be used as a stand-alone device or readily interfaced with existing medical instruments—for example, introduced through the working channel of a standard gastroscope for imaging internal organs. In contrast to the current standard of care—tissue biopsy followed by *ex vivo* histopathology, which is performed more than 55 million times annually [1], 2PEM enables **label-free, real-time histology *in vivo* without tissue removal**. As a result, imaging can be performed over a large area multiple times, thereby reducing the high sampling			

	error inherent to conventional biopsy procedures. Beyond structural information, 2PEM can also provide **functional insights** into biological tissues, such as cellular metabolism and redox state, which are not accessible with standard histopathology. These unique capabilities position 2PEM to potentially drive a **paradigm shift in diagnosis**. Moreover, the availability of real-time histological information makes 2PEM particularly valuable for **intraoperative surgical guidance**. Collectively, the potential applications, clinical impact, and market opportunity of 2PEM are substantial. **Potential Use:** The proposed 2PEM technology is primarily intended for **clinical applications** with several medical uses explicitly outlined below: • ***In vivo* tissue optical histology**: Initially 2PEM can be used to guide biopsy and improve diagnostic yield. Once its performance is validated through co-registered standard histology, 2PEM may enable immediate histology-like evaluation/diagnosis without excision. • **Intraoperative guidance**: 2PEM provides real-time margin assessment during cancer surgery to support more complete tumor resection. This approach addresses key limitations of intraoperative frozen-section histology, including procedural delay/holdoff, restricted sampling sites/areas, and very limited repeatability. • **Longitudinal studies:** The technology enables repeated imaging of the same tissue over time, allowing tracking disease progression or treatment response. In addition to clinical use, 2PEM serves as a **powerful tool for basic and translational research**. Its compact footprint (2–4 mm in diameter) and ultralight weight (a few grams) enable applications that are impractical for conventional benchtop microscopes. Examples include imaging functional brain circuits in freely-behaving rodents and deployment within incubators for repeated, nondestructive monitoring of cells, engineered tissues, or organoids. Furthermore, 2PEM can be **highly cost-effective for infectious disease research or field use**, as only the inexpensive fiber-optic probe needs to be in direct contact with infectious materials and can be treated as disposable. This minimizes the risk of contaminating costly components of the imaging system (such as microscopes or laser systems) while maintaining high-resolution imaging capability.
Rationale for Joint R&D Project	**Technology Characteristics and Multidisciplinary Requirements:** The proposed 3D two-photon endomicroscopy (2PEM) technology is a highly integrated platform combining optics, precision opto-mechanics, piezoelectric actuation, microfabrication, electronics, control software, system packaging, and biomedical validation. Due to this vertical integration and high engineering complexity, the technology cannot be effectively developed or scaled by a single organization. Joint R&D is required to integrate academic strengths in optical physics, system design, and biomedical validation with industrial capabilities in precision manufacturing, packaging, reproducibility, and mass production. **State of the Art and Technical Gaps:** All-fiber scanning 2PEM has demonstrated unique capabilities for real-time, label-free *in vivo* optical histology and high-resolution biological imaging [2]. However, current systems remain at the laboratory prototype stage and face key limitations that prevent industrialization and clinical deployment. These include (1) the lack of compact, high-precision depth-scanning for true 3D imaging; (2) dependence on limited global suppliers of tubular PZT actuators with suboptimal form factors and performance consistency; and (3) manual, low-yield assembly of the scanning probe head, resulting in poor reproducibility and limited scalability. **Need for Collaboration with Korean Industry/Research Institution:** Korea has a well-established SME industrial ecosystem for precision opto-mechanical and electro-mechanical components, including miniature piezoelectric devices, micro-optics, miniature linear scanners, and advanced packaging technologies. These capabilities are essential for overcoming current technical bottlenecks, enabling reproducible manufacturing, improving yield, and supporting cost reduction and potential disposability of the 2PEM probe. Collaboration with Korean SMEs and/or research institutions is therefore critical to transition 3D-2PEM from a laboratory system to a commercially

	viable platform. Successful development is expected to create significant industrial value and open domestic and global markets in biomedical imaging and advanced medical devices.
Key Words	1. 3D Two-photon Endomicroscope (3D-2PEM) 2. Label-free *in vivo* Optical Histology 3. Miniaturized Depth-Scanning Mechanisms (Micro Mechanics or Focus-Tunable Micro-optics) 4. Compact Tubular Piezoelectric Scanners 5. Ultracompact Fiber-Optic Probe Packaging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	**Overall Objective:** To develop a commercially deployable, ultracompact fiber-optic 3D two-photon endomicroscopy (3D-2PEM) platform—including scanning probe, control hardware, and software—capable of real-time, label-free, *in vivo* subcellular-resolution imaging, and to advance the technology from TRL 3 to TRL 6. **Specific Objectives:** • Develop a **miniaturized depth-scanning module**, based on a micro-mechanical device or focus-tunable micro-optic, suitable for ultracompact probes. • Integrate depth-scanning with the **fiber-optic 2PEM probe** for true 3D imaging. • Develop **synchronized control hardware and software** that coordinates depth scanning with existing 2D beam scanning in the fiber probe. • Design and fabricate **compact tubular piezoelectric actuators** with optimized dimensions and electroless nickel surface electrodes. • Establish **reliable electrical interconnection processes** (e.g., micro-welding) for attaching ultra-fine drive wires to piezo actuator electrodes. • Develop **modular, high-precision assembly and packaging processes** for the 2PEM scanning head, ensuring high reproducibility and component reusability. • **Characterize and optimize system performance** through iterative design, including 3D imaging of phantoms and biological tissues with validation against standard histology.			
R&D Target Specifications				

Category	Target Specification (quantitative metrics)
1. Miniature Depth Scanner	Diameter ≤ 4mm and length ≤ 10 mm; Scanning range ≥ 300 um, precision ≤1 um, repeatability ≤ 5 um, speed ≥10 um/sec.
2. 2PEM probe capable of depth scanning	Probe diameter at distal end ≤ 2.5 mm, at the depth scanner ≤ 4 mm, rigid length ≤ 60 mm.
3. Control hardware and software	Command execution delay ≤ 1ms; simple user interface for scanning range, speed, and step (or equivalent).
4. Compact tubular piezoelectric actuators	OD ≤ 1.5 mm, ID ≥ 0.75mm, length ≥15 mm, wall thickness ≤ 300 um, nickel electrode on its outer surface (through electroless deposition); actuation frequency ≥ 5 kHz; actuation voltage limit ≥ +/- 80V
5. Electrical interconnection processes	Connection of 50-100 um metal wires on the piezo tube surface with a soldering spot thickness ≤ 250 um (and a lateral size ≤500 um) with excellent conductivity while avoiding short circuit

6. Modular assembly and packaging processes	Concentricity and longitudinal tolerance ≤ 100 um
7. System performance	Imaging resolution: 1 um (lateral) x 15 um (depth); speed: 5 frames/sec; image size: 300 (lateral) x 300 um (depth); Probe size: 2.5-4 mm in diameter, rigid length < 60 mm

R&D Plan

Phase 1: Miniature Depth Scanner and Piezoelectric Actuator Development (Months 1–12)

- Systematically evaluate feasible depth-scanning approaches, including mechanical scanners with moving components and focus-tunable miniature lenses, with consideration of engineering complexity, technical risk, manufacturability, and cost; identify primary and backup solutions
- Design and fabricate the selected miniature depth-scanning mechanism
- Investigate fabrication processes for tubular piezoelectric actuators, including tube pulling and electrode deposition
- Design and fabricate tubular piezoelectric actuators compatible with the miniature scanning architecture
- Quantitatively characterize depth-scanner performance, including scanning range, accuracy, speed, and repeatability
- Characterize piezoelectric actuator performance, including actuation frequency, displacement, and voltage limits
- Iteratively optimize the scanner design and piezoelectric fabrication process based on experimental feedback
- Conduct a Phase-1 technical review to assess readiness for Phase-2 development against predefined quantitative performance metrics (see previous section)

Phase 2: Development of 3D 2PEM Probes with a Modular, Scalable Scanning Head Design (Months 13–36)

- Design and implement a modular scanning-head architecture for the 2PEM probe that enables rapid, reproducible assembly and packaging, supports component reuse, and is compatible with scaled-up manufacturing
- Develop fully functional 3D 2PEM probes by integrating the miniature depth scanner with the tubular piezoelectric actuator
- Integrate depth-scanning control with existing 2D fiber-optic beam scanning and image-acquisition hardware/software to enable synchronized 3D imaging
- Characterize and optimize 3D imaging performance using tissue-mimicking phantoms, refining system design and fabrication processes to meet target specifications defined in the prior section
- Validate system performance in biological tissues and relevant animal models for label-free, *in vivo* optical histology, with direct comparison to standard histopathology
- Complete system packaging and integration, and demonstrate performance consistent with **TRL-6 readiness**

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The international research institute will provide the design of the state-of-the-art baseline technology platform and lead efforts on scientific feasibility, system validation, and technical risk reduction. Specific responsibilities include: • Providing research-grade 2PEM system concepts and core system-level design expertise • Establishing a baseline 2PEM probe capable of high-quality 2D optical histology imaging • Conducting proof-of-concept optical histology studies using biological tissues and appropriate *in vivo* animal models • Validating 3D 2PEM system performance for optical histology through direct comparison with co-registered standard histopathology on identical samples • Participating in phase-gate technical reviews and contributing to the definition of quantitative performance benchmarks • Supporting knowledge transfer and technical handoff to enable transition from a research-grade prototype to a commercial-grade, manufacturable platform
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the lead industrial entity, responsible for product-oriented system development, manufacturing readiness, and commercialization planning. Specific responsibilities include: • Defining the overall system architecture and providing leadership for product-focused development • Designing and fabricating the miniature depth scanner to commercial-grade standards • Developing commercial-grade control hardware and software for the depth-scanning module • Designing and fabricating tubular piezoelectric actuators suitable for reliable, scalable production • Developing a modular scanning-head architecture for the 3D-2PEM fiber-optic probe • Assembling fully integrated 3D 2PEM probes and delivering systems ready for characterization and performance validation on biological tissues and animal models by the international research institute • Performing system-level optimization and integration to support technology readiness level (TRL) advancement • Leading project management activities, intellectual property (IP) strategy, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**General Outcomes and Impacts:** At the conclusion of this program, we will deliver an enabling technology platform that is well positioned for establishing a new paradigm for medical diagnosis and intra-operative guidance by providing 3D, label-free "optical histology" of biological tissues *in vivo* and in real time, without the need for tissue removal. Key engineering barriers related to miniaturization, reliability, and manufacturability will be overcome, enabling scalable, reproducible production and readiness for commercialization. Beyond clinical translation—which represents a substantial and growing market—this leading-edge technology platform will also enable a wide range of discovery-driven research directions. As a result, the anticipated impact is broad, cross-sectoral, and long-lasting. **Technological Outcomes** • Establishment of an ultracompact fiber-optic imaging platform capable of **high-resolution, real-time, label-free three-dimensional (3D) imaging** • Advancement of **miniature precision scanning technologies and/or tunable micro-optics**, with applicability beyond the proposed platform • Establishment of **manufacturing-ready processes** for high-quality miniature tubular piezoelectric actuators, addressing global demand and enabling use across multiple application domains • Development of **robust packaging and assembly protocols** for miniature, high-performance imaging devices with excellent scalability and reproducibility **Commercial and Industrial Impact** • Development of an enabling imaging technology platform addressing a **large and unmet clinical market** • Realization of **disposable 3D-2PEM probes**, significantly enhancing the value proposition, clinical adoption potential, and regulatory pathway • Commercialization potential extending beyond integrated 3D-2PEM probes, as key components developed in this project—such as miniature precision linear scanner or tunable micro-optics, and tubular piezoelectric actuators—can be deployed across **multiple MedTech, industrial, and research applications** **Broader Impact** • Enabling a **cost-effective paradigm shift** in clinical diagnosis and interventional guidance by **addressing major limitations** of current standards of care, including sampling errors, diagnostic delays, and tissue removal associated with biopsy • Strengthening **Korea's global competitiveness** in MedTech, next-generation diagnostic tools, and advanced *in vivo* imaging platforms, with spillover benefits to basic research and materials inspection • Enhancing opportunities for **international collaboration, technology transfer, and export-oriented industrial growth**

6. References

[1] https://www.researchnester.com, on biopsy device markets.

[2] W. X. Liang, G. Hall, B. Messerschmidt, M-J Li, and X. D. Li, "Nonlinear optical endomicroscopy for label-free functional histology in vivo," ***Light: Science and Applications*** 6:e17082 (2017).